

Problem 29.30

$$8.6 \times 10^{13} \text{ J/kg} \times 235 \times u \times n = 1.0 \text{ J/s}$$

**Combustion value
of U fission**

$$u = 1.66 \times 10^{-27} \text{ kg}$$

$$n = 3.0 \times 10^{10} / \text{s}$$

Problem 29.33

$$R = 1.2 \times 10^{-15} \text{ m} \times 7^{1/3} = 2.3 \times 10^{-15} \text{ m}$$

$$2E_k = \frac{1}{4\pi\epsilon_0} \frac{(3e)^2}{2R}$$

$$E_k = \frac{9}{4} \left(\frac{1}{4\pi\epsilon_0} \frac{e^2}{2a_0} \right) \frac{2 \times 0.0529 \text{ nm}}{2.3 \times 10^{-6} \text{ nm}}$$
$$= 1.4 \text{ MeV}$$

Problem 30.4

$$\mu^{-} + p \rightarrow n + \nu_{\mu}$$

Problem 30.5

(a) $\Lambda^0 \rightarrow \pi^+ + \pi^-$ B, spin

(b) $K^+ \rightarrow \pi^+ + \pi^- + \pi^0$ Q

(c) $\bar{n} \rightarrow e^- + p + \bar{\nu}_e$ B

(d) $n \rightarrow e^- + e^+ + \nu_e$ B L_e

Λ is strange particle, $S = -1$.

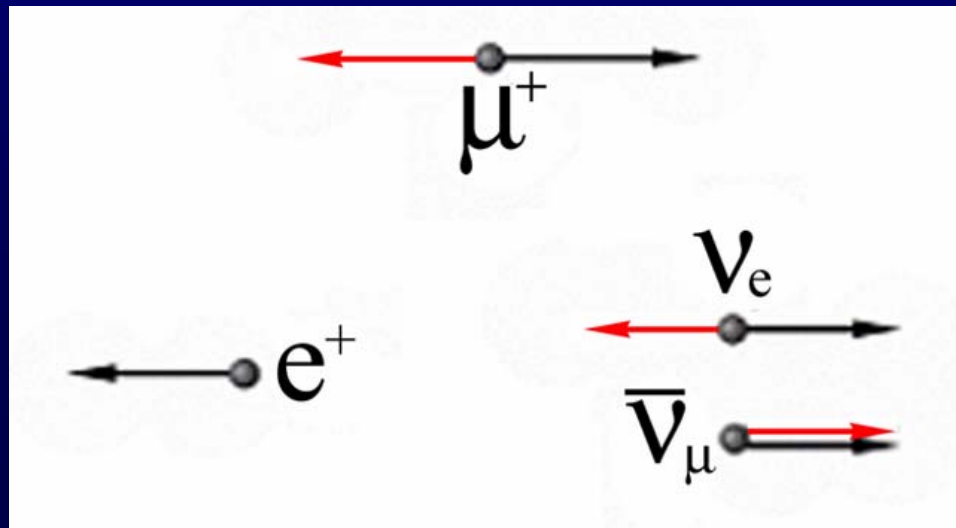
(a) is a weak process. ($\Delta S = \pm 1$)

EM & strong interactions conserve S.

Problem 30.9

$$\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$$

Since the positive muon is left-handed. The direction of its spin is backward. Because the neutrino is left-handed while the anti-neutrino is right-handed and the two neutrinos both move forward, the two spin angular-momenta cancel each other. Then the direction of the spin of the positron must be backward due to the conservation of the angular momentum. Therefore, the positron is right-handed.



Problem 30.10 Suppose the tau neutrino exists and sometimes a tau lepton can decay into three muons and two neutrinos. Specify the two neutrinos.

$$\tau^- \rightarrow \mu^- + \mu^+ + \mu^- + \nu_\tau + \bar{\nu}_\mu$$

$$L_\tau = 1 \qquad 1$$

$$L_\mu = 1 \qquad 1 \quad -1 \quad 1 \quad -1$$

$$Q = -1 \qquad -1 \quad 1 \quad -1$$